

# LAB 2 REPORT

## Agile Backlog Creation & Sprint Simulation in Jira

### Project: Smart Parking Space Rental & License Validator

|                     |                                                |
|---------------------|------------------------------------------------|
| <b>Student Name</b> | Prakash Heroor                                 |
| <b>Project</b>      | Smart Parking Space Rental & License Validator |
| <b>Domain</b>       | Smart Cities, Transport & Logistics            |
| <b>Users</b>        | Vehicle Owner, Parking Warden                  |

### Objective

- To convert functional requirements into Agile backlog items.
- Organize the requirements into an Epic and User Stories.
- Prioritize and estimate the User Stories.
- Simulate a Scrum sprint in Jira.
- Analyze sprint progress using the Burndown Chart.

### 2. Epic

|                    |                                                                                                    |
|--------------------|----------------------------------------------------------------------------------------------------|
| <b>Epic Name</b>   | Smart Parking Space Management                                                                     |
| <b>Description</b> | Enable vehicle owners to easily find, rent, and manage available parking spaces through a smart pa |

### 3. User Stories, Priority & Story Points

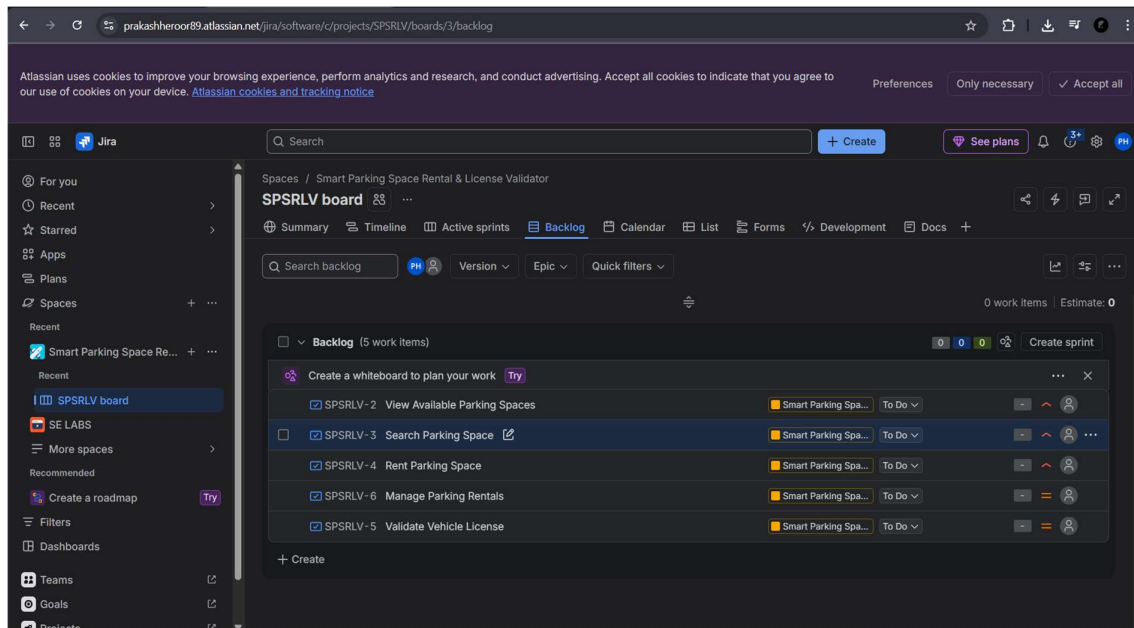
The User Stories follow the format: As a [user role], I want [goal], so that [benefit].

| Issue    | Story                         | User           | User Story                                                                                                                                 | Priority | Points |
|----------|-------------------------------|----------------|--------------------------------------------------------------------------------------------------------------------------------------------|----------|--------|
| SPSRLV-2 | View Available Parking Spaces | Vehicle Owner  | As a vehicle owner, I want to view available parking spaces, so that I can find a suitable parking spot.                                   | High     | 3      |
| SPSRLV-3 | Search Parking Space          | Vehicle Owner  | As a vehicle owner, I want to search for parking spaces based on location and availability, so that I can quickly find convenient parking. | High     | 5      |
| SPSRLV-4 | Rent Parking Space            | Vehicle Owner  | As a vehicle owner, I want to rent an available parking space, so that I can reserve a parking spot for my vehicle.                        | High     | 8      |
| SPSRLV-5 | Validate Vehicle License      | Parking Warden | As a parking warden, I want to validate a vehicle's license, so that I can ensure that only authorized vehicles use the parking facility.  | Medium   | 5      |
| SPSRLV-6 | Manage Parking Rentals        | Vehicle Owner  | As a vehicle owner, I want to view and manage my parking rentals, so that I can keep track of my reserved parking spaces.                  | Medium   | 3      |

**Total Story Points: 24**

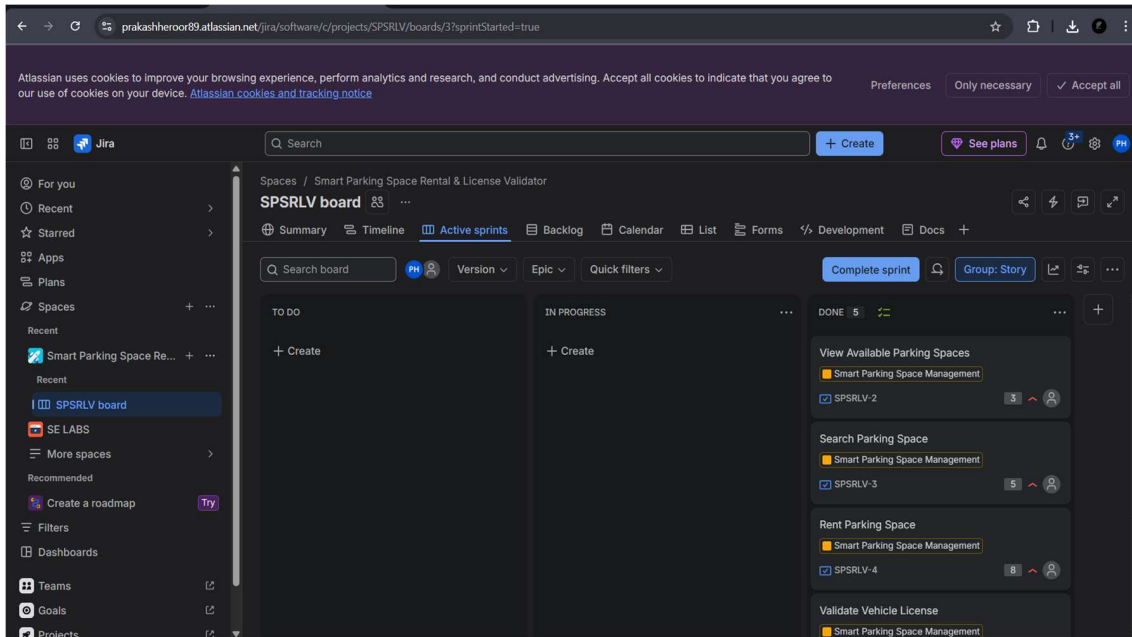
### 4. Jira Backlog

- The Jira backlog contains the five User Stories under the Smart Parking Space Management Epic.
- The backlog also displays the assigned Story Points.



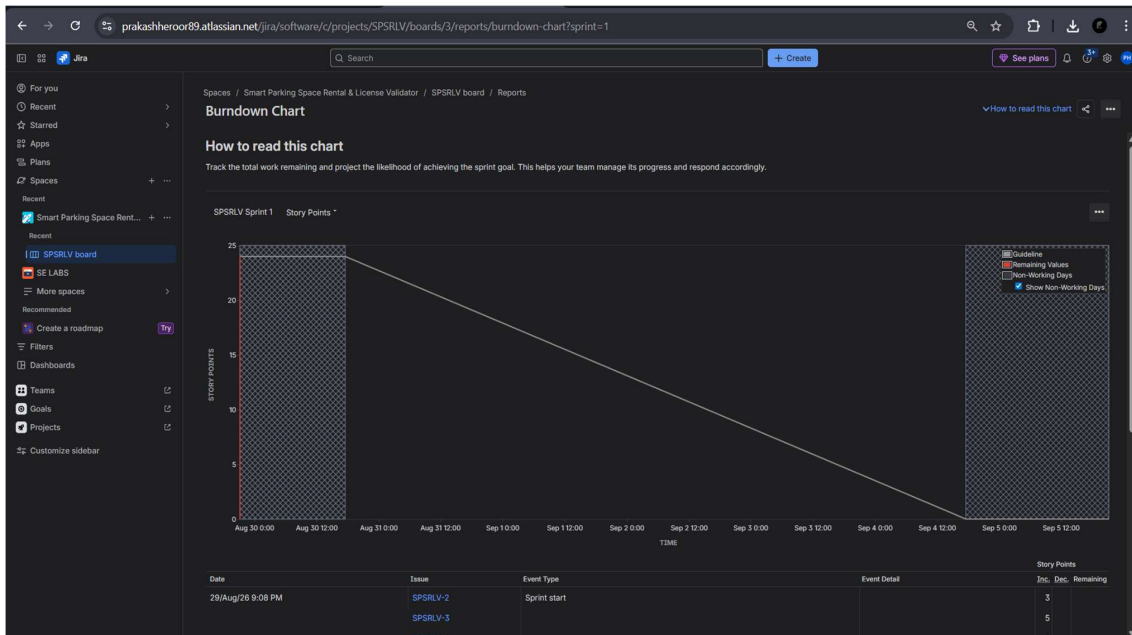
## 5. Sprint 1 Simulation

- Sprint 1 was created with the five User Stories selected for the sprint.
- The sprint was planned as a one-week sprint.
- The stories were moved through the Scrum workflow: To Do → In Progress → Done.



## 6. Burndown Chart

- The Burndown Chart tracks the amount of remaining work during Sprint 1.
- Story Points are used as the estimation statistic.
- The sprint started with 24 Story Points.
- The guideline shows the expected linear reduction of work over the sprint.



## **7. Reflection Questions**

### **1. Did your estimations reflect the actual effort?**

- The estimations provided a reasonable relative measure of effort for the simulated sprint. The larger story, Rent Parking Space, was assigned 8 points because it involves more complexity than the other stories. Since this was a simulated sprint, the story points mainly represent relative complexity and uncertainty rather than exact development time.

### **2. Was your backlog well-prioritized?**

- Yes. The core parking functions were given High priority because they directly support the main objective of the system. View Available Parking Spaces, Search Parking Space, and Rent Parking Space were prioritized first. License validation and rental management were given Medium priority.

### **3. How did your simulated sprint align with your plan?**

- The sprint followed the planned Scrum workflow. The selected stories were added to Sprint 1 and progressed through the board. The simulation demonstrated the movement of work from To Do to In Progress and finally to Done.

### **4. What insights did the burndown chart give about your team's capacity?**

- The Burndown Chart showed the planned reduction of the 24 story points across the sprint. It provides a visual indication of remaining work and helps compare actual progress with the expected guideline. In this simulation, it also demonstrates how Jira can be used to monitor sprint progress and assess whether the team is on track.